PRACTICAL 04 CLOUD COMPUTING

DT :13-01-2026

API KEY: AIzaSyAOVYRIgupAurZup5y1PRh8Ismb1A3lLao

FILENAME: mapservlet:

```
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/mapservlet")
public class mapservlet extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String lat = request.getParameter("lat");
        String lng = request.getParameter("lng");

        // Basic validation
        try {
            Double.parseDouble(lat);
            Double.parseDouble(lng);
        } catch (NumberFormatException e) {
            response.getWriter().println("Invalid coordinates. Please enter valid numbers.");
            return;
        }

        request.setAttribute("lat", lat);
```

```
        request.setAttribute("lng", lng);
        request.getRequestDispatcher("map.jsp").forward(request, response);
    }
}
```

FILENAME: index.HTML

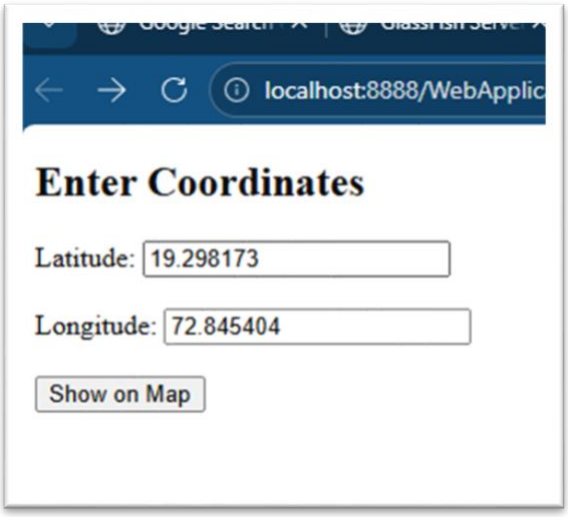
```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head>
    <title>Google Maps Location Finder</title>
</head>
<body>
<h2>Enter Coordinates</h2>
<form action="mapservlet" method="get">
    Latitude: <input type="text" name="lat" required><br><br>
    Longitude: <input type="text" name="lng" required><br><br>
    <input type="submit" value="Show on Map">
</form>
</body>
</html>
```

FILENAME: map.jsp

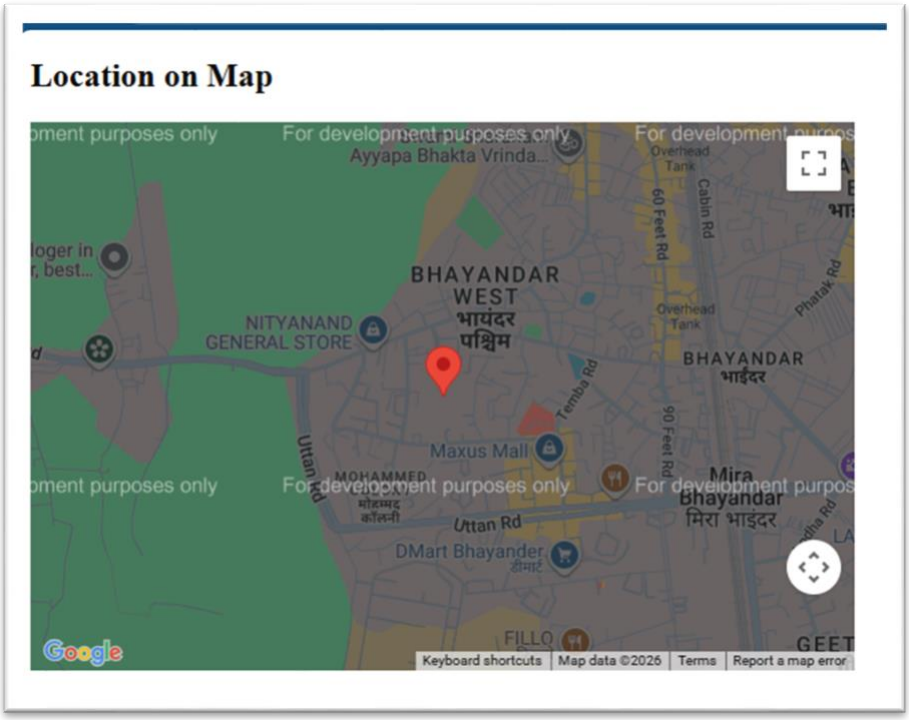
```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%
    String lat = (String) request.getAttribute("lat");
    String lng = (String) request.getAttribute("lng");
    String apiKey = "AIzaSyAOVYRIgupAurZup5y1PRh8Ismb1A3lLao"; // Replace with your API key
%>
<!DOCTYPE html>
<html>
<head>
    <title>Map Location</title>
    <script src="https://maps.googleapis.com/maps/api/js?key=<%= apiKey %>"></script>
    <script>
        function initMap() {
            var location = {lat: parseFloat("<%= lat %>"), lng: parseFloat("<%= lng %>")};
            var map = new google.maps.Map(document.getElementById("map"), {
                zoom: 15,
                center: location
            });
            new google.maps.Marker({
                position: location,
                map: map
            });
        }
    </script>
</head>
<body onload="initMap()">
```

```
<h2>Location on Map</h2>
<div id="map" style="width: 600px; height: 400px;"></div>
</body>
</html>
```

OUTPUT



4.1



4.2